

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

Nov - Dec 2013

EE310 Programmable Logic Controller and Industrial Automation

Date: 07.12.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

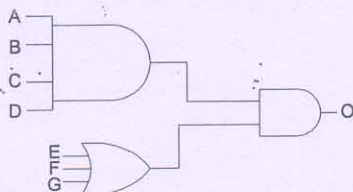
- Q - 1 (a) State any seven advantages of PLC [07]
Q - 2 (a) State the difference between sequential/ logic control and analog/automatic control [07]
(b) Explain input module of PLC [07]

OR

- Q - 2 Prepare ladder logic for following gate logic AND, OR, NOT, NOR, NAND, EXOR [14]
and EXNOR

- Q - 3 (a) Draw symbols for following: [05]
1. Normally open contact
2. Normally closed contact
3. Latch Contact
4. Unlatch Contact
5. Output coil

- (b) Draw ladder logic for the following: [09]
1. For output 7 to be on, input 6 must be on and either input 8 or input 9 must be on. In addition one of input 1,2 or 3 must be on.
2. For output H to be on, input A must be on and both inputs C and D must be off. In addition one or more inputs E, F and G must be off.
3.



OR

- Q - 3 List out types of counters and draw and explain function block of each type, also stating application of each type [14]

SECTION - II

- Q - 4 (a) What are the factors to be considered while selecting a PLC? Explain each one in short [07]
Q - 5 List out types of timers and draw and explain function block of each type, also stating application of each type [14]

OR

- Q - 5 (a) Prepare ladder logic for the following: There are 3 mixing devices on a processing line: A, B, C. After the process begins, mixer A is to start after 7s elapse. Next mixer B is to start 3.6s after A. Mixer C is to start 5s after B. All then remain on until a master enable switch is turned off. [07]

- (b) Draw and explain PLC scanning process [07]
- Q - 6 (a) Prepare ladder logic for 4 motors M1, M2, M3 and M4 are to turn ON in sequential way one after other after some time interval. In this process, the sequence should not stop at M4 but the motor M1 starts once again (after M4 off) and sequence run continuously. [07]
- (b) Define the following: [07]
1. Shared Controller
 2. Relay
 3. Control Valve
 4. Process
 5. PLC
 6. Protocol
 7. Transducer

OR

- Q - 6 (a) Prepare a ladder for a solenoid S to go on when count C goes up to 22 and when count D goes down from 37 to 0 and when count E goes up to 8. Furthermore if count F goes down from 17 to 0 at any time the solenoid is to be kept from operating. One stop button resets the entire process, including the solenoid's being off. [07]
- (b) Draw and explain block diagram of PLC [07]
